

# Generalized Lifting and Projection Theorems for Cylindrical Algebraic Decomposition

Generalization of McCallum (1985), Chapter 3

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## Abstract

We state and prove generalizations of McCallum’s Lifting Theorem (3.2.1) and Projection Theorem (3.2.3). The generalized versions replace the discriminant  $D(x) = \text{disc}(f)$  and specific resultants  $\text{res}(F_i, F_j)$  by arbitrary nonzero elements of the corresponding elimination ideals  $\langle f, \partial f / \partial x_r \rangle \cap \mathbb{R}[x]$  and  $\langle F_i, F_j \rangle \cap \mathbb{R}[x]$ . The proofs follow the structure of McCallum’s original argument with two new ingredients: (1) the *order additivity lemma* for holomorphic functions, and (2) a *norm identity*: for a Weierstrass polynomial  $h$  of degree  $m$ , any  $P \in \langle h, h' \rangle \cap \mathcal{O}(\Delta)$  satisfies  $P^m = \pm \text{disc}(h) \cdot Q$  for some holomorphic  $Q$ .

Sections that are unchanged from the original thesis are marked **[UNCHANGED]** and only summarized. Sections that require modification are marked **[CHANGED]** and written out in full.

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## 1 Setup and notation

[UNCHANGED] All notation, definitions, and background from Chapter 2 of the thesis carry over without modification. In particular:

- $K = \mathbb{R}$  or  $\mathbb{C}$ ; analytic/holomorphic functions; power series (Sec. 2.1).
- Submanifolds, coordinate systems, charts (Sec. 2.2, Theorems 2.2.1–2.2.4).
- Order  $\text{ord}_p f$ ; order-invariance; Theorem 2.3.1 (order under composition); root continuity principle (Theorem 2.3.2).
- Resultant, discriminant, Theorem 2.3.3 (discriminant of a product).
- Weierstrass preparation theorem (Theorem 2.1.10).
- Zariski’s 1975 theorem (Theorem 4.1.1) and its proof (Chapter 4).

Throughout,  $x = (x_1, \dots, x_{r-1})$  and  $(x, x_r)$  denotes an  $r$ -tuple. A polynomial  $f(x, x_r) \in \mathbb{R}[x, x_r]$  is viewed as univariate in  $x_r$  over  $\mathbb{R}[x]$ .

## 2 The generalized projection map $P'(A)$

[CHANGED]

**Definition 2.1** (Squarefree basis). A finite set  $A$  of polynomials in  $\mathbb{R}[x, x_r]$  is a *squarefree basis* if every element has positive degree in  $x_r$ , every element is squarefree (in  $\mathbb{R}[x, x_r]$ ), and the elements are pairwise coprime.

**Definition 2.2** (Generalized reduced projection). Let  $A$  be a squarefree basis. The *generalized reduced projection*  $P'(A)$  is a set of polynomials in  $\mathbb{R}[x]$  of the form

$$P'(A) = \text{coeff}(A) \cup \text{elim\_self}(A) \cup \text{elim\_pair}(A),$$

where:

- $\text{coeff}(A) = \{c \in \mathbb{R}[x] : c \text{ is a nonzero coefficient of some } F \in A \text{ viewed as a polynomial in } x_r\}$ .
- $\text{elim\_self}(A) = \{d_F : F \in A, \deg(F) \geq 2\}$ , where each  $d_F$  is a *chosen* nonzero element of  $\langle F, F' \rangle \cap \mathbb{R}[x]$ . (Here  $F' = \partial F / \partial x_r$ .)
- $\text{elim\_pair}(A) = \{r_{F,G} : F, G \in A, F \neq G, \deg(F) \geq 1, \deg(G) \geq 1\}$ , where each  $r_{F,G}$  is a *chosen* nonzero element of  $\langle F, G \rangle \cap \mathbb{R}[x]$ .

**Remark.** Taking  $d_F = \text{disc}(F)$  and  $r_{F,G} = \text{res}(F, G)$  recovers the original projection  $P(A) = \text{coeff}(A) \cup \text{disc}(A) \cup \text{res}(A)$ . The discriminant and resultant are particular nonzero elements of the respective elimination ideals (see Lemma 2.3).

**Lemma 2.3.** *Let  $F \in \mathbb{R}[x, x_r]$  be squarefree of degree  $\geq 2$  in  $x_r$ . Then  $\text{disc}(F) \neq 0$  and  $\text{disc}(F) \in \langle F, F' \rangle \cap \mathbb{R}[x]$  (up to a unit factor involving the leading coefficient).*

*Similarly, for  $F \neq G$  coprime in  $\mathbb{R}[x, x_r]$ ,  $\text{res}(F, G) \neq 0$  and  $\text{res}(F, G) \in \langle F, G \rangle \cap \mathbb{R}[x]$ .*

*Proof.* Standard properties of resultants and discriminants:  $\text{res}(F, G) \in \langle F, G \rangle$  follows from the Sylvester matrix identity, and  $\text{disc}(F) = \pm \text{res}(F, F')/\text{lc}(F)$ .  $\square$

### 3 Prerequisite: order-invariance of products (Lemma 3.2.2)

[UNCHANGED]

**Lemma 3.1** (Lemma 3.2.2). *Let  $K = \mathbb{R}$  or  $\mathbb{C}$ . Let  $F_1, \dots, F_n$  be nonconstant analytic functions defined in an open subset  $U$  of  $K^r$ , and let  $f = F_1 \cdots F_n$ . Let  $S$  be a connected subset of  $U$ . Then  $f$  is order-invariant in  $S$  if and only if each  $F_i$  is order-invariant in  $S$ .*

*Proof.* By additivity of the vanishing order:  $\text{ord}_p(f) = \sum_i \text{ord}_p(F_i)$ .  $\square$

### 4 New prerequisite: order additivity lemma

[NEW]

**Lemma 4.1** (Order additivity lemma). *Let  $U \subseteq \mathbb{C}^s$  be a connected open set. Let  $f, g : U \rightarrow \mathbb{C}$  be holomorphic functions, both not identically zero. If  $f \cdot g$  has constant vanishing order on  $U$  (i.e.,  $\text{ord}_z(fg)$  is the same for all  $z \in U$ ), then  $f$  and  $g$  each have constant vanishing order on  $U$ .*

*Proof.* Let  $c = \text{ord}_z(fg)$  (constant) for all  $z \in U$ . Let  $a = \min_{z \in U} \text{ord}_z(f)$  and  $b = \min_{z \in U} \text{ord}_z(g)$ .

Since  $f$  is holomorphic and not identically zero, the set  $\{z \in U : \text{ord}_z(f) \geq k\}$  is a closed analytic subset of  $U$  for each  $k$ . The order achieves its minimum  $a$  on a dense open subset  $U_1 \subseteq U$  (the complement of a proper analytic subset). Similarly,  $\text{ord}_z(g)$  achieves its minimum  $b$  on a dense open  $U_2 \subseteq U$ .

On  $U_1 \cap U_2$  (which is dense and open, since  $U$  is connected and each  $U_i$  is the complement of a proper analytic subset), we have  $a + b = c$ .

For an arbitrary point  $z_0 \in U$ :

- $\text{ord}_{z_0}(f) \geq a$  (since  $a$  is the minimum),
- $\text{ord}_{z_0}(g) \geq b$  (since  $b$  is the minimum),
- $\text{ord}_{z_0}(f) + \text{ord}_{z_0}(g) = c = a + b$ .

The third equality together with the two inequalities forces  $\text{ord}_{z_0}(f) = a$  and  $\text{ord}_{z_0}(g) = b$ .  $\square$

**Remark.** This lemma generalizes to any connected analytic manifold, and also to the real-analytic setting (by complexification). The key input is that for a non-identically-zero holomorphic function on a connected open set, the vanishing order achieves a finite minimum on a dense open subset and can only jump up on a proper analytic subset.

## 5 New prerequisite: norm identity for monic polynomials

[NEW]

**Remark.** An earlier version of this document claimed that  $\langle h, g \rangle \cap R = (\text{res}_{z_r}(h, g))$  for  $h$  monic over an integral domain  $R$ . This is **false** in general: for  $R = \mathbb{Q}[t]$ ,  $h = z_r^3 + t$ ,  $g = 3z_r^2$ , the element  $3t \in \langle h, g \rangle \cap R$  but  $\text{res}(h, g) = 27t^2 \nmid 3t$  in  $\mathbb{Q}[t]$ . The correct tool is the norm identity below.

**Lemma 5.1** (Norm identity for monic polynomials). *Let  $R$  be a commutative ring, and let  $h \in R[z_r]$  be monic of degree  $m$ . Let  $g \in R[z_r]$ . If  $P \in R$  satisfies  $P \in \langle h, g \rangle \cap R$  (i.e.,  $C(P) = h \cdot a + g \cdot b$  for some  $a, b \in R[z_r]$ ), then*

$$P^m = (-1)^{m \deg g} \text{res}_{z_r}(h, g) \cdot N(\bar{b}),$$

where  $\bar{b}$  is the image of  $b$  in the free  $R$ -module  $R[z_r]/(h)$  and  $N = N_{R[z_r]/(h)/R}$  is the algebra norm (determinant of the multiplication map).

*Proof.* Since  $h$  is monic of degree  $m$ , the quotient  $R[z_r]/(h)$  is a free  $R$ -module of rank  $m$  with basis  $\{1, \bar{z}_r, \dots, \bar{z}_r^{m-1}\}$ . Reducing the identity  $C(P) = h \cdot a + g \cdot b$  modulo  $h$  gives  $\bar{g} \cdot \bar{b} = P$  in  $R[z_r]/(h)$ .

The norm map  $N : R[z_r]/(h) \rightarrow R$  is the multiplicative map sending  $\bar{a}$  to the determinant of  $x \mapsto \bar{a} \cdot x$ . Applying  $N$  to both sides:

$$N(\bar{g}) \cdot N(\bar{b}) = N(P) = P^m,$$

using multiplicativity of the norm and the fact that  $N$  of a scalar  $P \in R \hookrightarrow R[z_r]/(h)$  equals  $P^m$  (since multiplication by  $P$  acts as  $P \cdot \text{Id}$  on an  $m$ -dimensional free module).

For  $h$  monic, the norm of  $\bar{g}$  in  $R[z_r]/(h)$  equals  $(-1)^{m \deg g} \text{res}_{z_r}(h, g)$  (this is the standard identity relating the algebra norm to the resultant; see Lang, *Algebra*, Ch. IV, §8, or Eisenbud, *Commutative Algebra*, Exercise 14.14).  $\square$

**Corollary 5.2.** *Let  $h(z, z_r) \in \mathcal{O}(\Delta)[z_r]$  be a Weierstrass polynomial of degree  $m$  in a polydisc  $\Delta \subseteq \mathbb{C}^{n-1}$ , and let  $h' = \partial h / \partial z_r$ . If  $P \in \langle h, h' \rangle \cap \mathcal{O}(\Delta)$ , then*

$$P^m = \pm \text{disc}(h) \cdot Q \tag{1}$$

for some  $Q \in \mathcal{O}(\Delta)$ , namely  $Q = N(\bar{b})$  where  $\bar{b}$  is determined by a representation  $P = h \cdot a + h' \cdot b$  in  $\mathcal{O}(\Delta)[z_r]$ .

*Proof.* Lemma 5.1 with  $g = h'$  gives  $P^m = (-1)^{m(m-1)} \text{res}_{z_r}(h, h') \cdot N(\bar{b})$ . Since  $h$  is monic,  $\text{disc}(h) = (-1)^{m(m-1)/2} \text{res}_{z_r}(h, h')$ , so  $\text{res}_{z_r}(h, h') = (-1)^{m(m-1)/2} \text{disc}(h)$ . Absorbing the sign into “ $\pm$ ” gives (1).  $\square$

## 6 The Generalized Lifting Theorem (Theorem 3.2.1')

[CHANGED: statement and proof]

**Theorem 6.1** (Generalized Lifting Theorem, Theorem 3.2.1'). *Let  $f(x, x_r) \in \mathbb{R}[x, x_r]$  be squarefree, of positive degree in  $x_r$ . Let  $S$  be a connected submanifold of  $\mathbb{R}^{r-1}$  in which  $f$  is degree-invariant and of non-negative degree. Suppose that some nonzero*

$$P \in \langle f, \partial f / \partial x_r \rangle \cap \mathbb{R}[x]$$

*is order-invariant in  $S$ . Then  $f$  is analytically delineable on  $S$  and is order-invariant in each  $f$ -section over  $S$ .*

**Remark.** This generalizes the original Theorem 3.2.1, which takes  $P = D(x) = \text{disc}(f)$ . Since  $\text{disc}(f) \in \langle f, f' \rangle \cap \mathbb{R}[x]$  (Lemma 2.3) and  $\text{disc}(f) \neq 0$  for squarefree  $f$ , the original theorem is a special case.

## 6.1 Proof structure

**[UNCHANGED]** The proof follows the same overall structure as the original. Let the dimension of  $S$  be  $s$ , where  $0 \leq s \leq r-1$ . If  $s = 0$  the theorem is trivial. For  $1 \leq s \leq r-1$ , the theorem follows from the following assertion about “local delineability”:

**Assertion** (Assertion 1). For each point  $p$  of  $S$  there is a neighborhood  $N \subseteq \mathbb{R}^{r-1}$  of  $p$  such that  $f$  is analytically delineable on  $S \cap N$  and  $f$  is order-invariant in each  $f$ -section over  $S \cap N$ .

**[UNCHANGED]** The deduction of the theorem from Assertion 1 is identical to the original: by connectedness of  $S$ , the number of distinct real roots and their multiplicities are locally constant, hence globally constant. The local analytic root functions patch to global root functions on  $S$ .

**[UNCHANGED]** It remains to establish Assertion 1. As in the original, we reduce to the case  $\alpha_i = 0$  (by translation in  $x_r$ ), fix  $m = m_i$ ,  $C = C_i$ , and seek a neighborhood  $N = N_i$  of  $p$  and an analytic function  $\theta : S \cap N \rightarrow (-\epsilon, +\epsilon)$  satisfying (3.3.1) and order-invariance.

We handle two cases.

## 6.2 Case $s = r - 1$ ( $S$ is open)

**[CHANGED — simplified]**

Since  $s = r - 1$ ,  $S$  is an open subset of  $\mathbb{R}^{r-1}$ . Let  $U \subseteq N_0$  be a neighborhood of  $p$  with  $U \subseteq S$ .

Since  $P$  is order-invariant on  $S$  and  $P \neq 0$ , and  $S$  is open and connected,  $P$  has constant finite order on  $S$ . On an open connected set, a holomorphic (indeed polynomial) function with constant vanishing order must be nowhere-vanishing (order 0) unless it is identically zero. Since  $P \neq 0$ , we have  $P(q) \neq 0$  for every  $q \in S$ .

Since  $P \in \langle f, f' \rangle \cap \mathbb{R}[x]$ , there exist  $A, B \in \mathbb{R}[x, x_r]$  such that

$$P(x) = A(x, x_r)f(x, x_r) + B(x, x_r)\frac{\partial f}{\partial x_r}(x, x_r).$$

If  $f(p, \alpha) = 0$  and  $\frac{\partial f}{\partial x_r}(p, \alpha) = 0$  for some  $\alpha$ , then evaluating the identity at  $(p, \alpha)$  gives  $P(p) = 0$ , contradicting  $P(p) \neq 0$ . Therefore  $f(p, x_r)$  and  $\frac{\partial f}{\partial x_r}(p, x_r)$  have no common root, i.e., all roots of  $f(p, x_r)$  are simple.

In particular,  $m = 1$  (the root  $\alpha_i = 0$  is simple), and  $\frac{\partial f}{\partial x_r}(p, 0) \neq 0$ . By the implicit function theorem (Theorem 2.1.7 with  $s = r - 1$ ), there exist neighborhoods  $N' \subseteq U$  and  $W \subseteq (-\epsilon, +\epsilon)$  of 0, and an analytic function  $\theta : S \cap N' \rightarrow W$  such that for all  $x \in S \cap N'$  and  $\alpha \in W$ ,

$$f(x, \alpha) = 0 \quad \text{if and only if} \quad \alpha = \theta(x).$$

Since  $m = 1$ ,  $\frac{\partial f}{\partial x_r}(x, \theta(x)) \neq 0$ , so  $\text{ord}_{(x, \theta(x))} f = 1$  for all  $x \in S \cap N'$ . Hence  $f$  is order-invariant (with constant order 1) in the graph of  $\theta$ . This establishes Assertion 2 (and hence Assertion 1) in this case.

**Remark.** Compare with the original proof (pp. 23–24 of the thesis), which uses the relation  $(-1)^{m(m-1)/2}R(x) = f_0(x)D(x)$  to conclude  $R(p) \neq 0$ . Our argument is simpler:  $P(p) \neq 0$  directly rules out a common root.

## 6.3 Case $1 \leq s \leq r - 2$ ( $S$ has positive codimension)

**[CHANGED — the factorization step is different; everything else is unchanged]**

### 6.3.1 Coordinate change (unchanged)

By Theorem 2.2.1, choose a coordinate system  $\Phi : U \rightarrow V$  about  $p$  with  $U \subseteq N_0$  connected, such that  $S \cap U = \{x \in U : \phi_{s+1}(x) = 0, \dots, \phi_{r-1}(x) = 0\}$ . Let  $T$  be the  $s$ -dimensional linear subspace  $\{y \in \mathbb{R}^{r-1} : y_{s+1} = \dots = y_{r-1} = 0\}$ , so the image of  $S \cap U$  under  $\Phi$  is  $T \cap V$ .

Define  $g(y, y_r)$  by transforming  $f(x, x_r)$  via  $\Phi$ :  $g_j(y) = f_j(\Phi^{-1}(y))$  for  $0 \leq j \leq d$ , so  $g(y, y_r) = \sum_{j=0}^d g_j(y) y_r^{d-j}$  is a pseudopolynomial with analytic coefficients.

### 6.3.2 Transform of $P$ (changed)

**[CHANGED]**

Define  $\tilde{P}(y) = P(\Phi^{-1}(y))$  for  $y \in V$ . Since  $P \in \mathbb{R}[x]$  and  $\Phi^{-1}$  is analytic,  $\tilde{P}$  is a real-analytic function in  $V$ , not identically zero (as  $P \neq 0$ ).

Since  $P(x) = A(x, x_r)f(x, x_r) + B(x, x_r)f'(x, x_r)$  in  $\mathbb{R}[x, x_r]$ , substituting  $x = \Phi^{-1}(y)$  gives

$$\tilde{P}(y) = \tilde{A}(y, y_r) g(y, y_r) + \tilde{B}(y, y_r) \frac{\partial g}{\partial y_r}(y, y_r)$$

where  $\tilde{A}(y, y_r) = A(\Phi^{-1}(y), y_r)$  and  $\tilde{B}(y, y_r) = B(\Phi^{-1}(y), y_r)$  are analytic (not polynomial) in  $(y, y_r)$ . Hence

$$\tilde{P} \in \langle g, \partial g / \partial y_r \rangle \cap \mathcal{O}(V),$$

where  $\mathcal{O}(V)$  denotes the ring of real-analytic functions of  $y$  in  $V$ .

Since  $P$  is order-invariant in  $S$  and  $\text{ord}_{G(p)} f \leq \text{ord}_p f \circ G$  (Theorem 2.3.1),  $\tilde{P}$  is order-invariant in  $T \cap V$ , with the same constant order as  $P$  on  $S$  (by the remark following the definition of order-invariance in Section 3.2 of the thesis).

### 6.3.3 Complexification and order-invariance on $T^*$ (unchanged in method)

**[UNCHANGED in method — the argument applies to  $\tilde{P}$  instead of the discriminant  $E$ ]**

Extend  $g(y, y_r)$  to a pseudopolynomial  $g(z, z_r)$  in a polydisc  $\Delta_1 \subseteq \mathbb{C}^{r-1}$  by extending each coefficient via its convergent power series (Theorem 2.1.2).

Similarly,  $\tilde{P}(y)$  extends to a holomorphic function  $\tilde{P}(z)$  in  $\Delta_1$ . The power series expansion of  $\tilde{P}(y)$  about 0 has real coefficients; substituting complex variables gives  $\tilde{P}(z)$ .

Let  $T^* = \{z \in \mathbb{C}^{r-1} : z_{s+1} = \dots = z_{r-1} = 0\}$  be the complexification of  $T$ . The order-invariance argument from the thesis (pp. 25–26) applies verbatim to  $\tilde{P}$ .

Let  $\mu = \text{ord}_0 \tilde{P}$ . Every partial derivative of  $\tilde{P}(z)$  of order  $< \mu$  vanishes throughout  $T^* \cap \Delta_1$  (because the corresponding real partial derivative vanishes on  $T \cap V$  by order-invariance, and the power series with real coefficients extends this to the complex domain). Some partial derivative of order  $\mu$  does not vanish at 0. After refining  $\Delta_1$ ,  $\tilde{P}(z)$  is order-invariant in  $T^* \cap \Delta_1$  with constant order  $\mu$ .

### 6.3.4 Weierstrass preparation (unchanged)

**[UNCHANGED]**

By the Weierstrass preparation theorem (Theorem 2.1.10), after refining  $\Delta_1$  to  $\Delta_2 \subseteq \Delta_1$ , there is a unit  $u(z, z_r)$  (holomorphic and non-vanishing in  $\Delta' = \Delta_2 \times \Delta(0; \epsilon)$ ) and a Weierstrass polynomial  $h(z, z_r) = z_r^m + a_1(z)z_r^{m-1} + \dots + a_m(z)$  in  $\Delta_2$  such that

$$g(z, z_r) = u(z, z_r) h(z, z_r) \tag{3.3.7}$$

for all  $(z, z_r) \in \Delta'$ , and for each fixed  $z \in \Delta_2$ , all roots of  $h(z, z_r)$  lie in  $\Delta(0; \epsilon)$ .

### 6.3.5 Recovering order-invariance of $\text{disc}(h)$ (changed — key new step)

**[CHANGED — this is the main new argument]**

In the original proof, the discriminant  $E(z) = \text{disc}(g)$  is decomposed as  $E(z) = Q(z) \cdot F(z)$  where  $F = \text{disc}(h)$  and  $Q$  involves  $\text{disc}(u)$  and  $\text{res}(u, h)$ , using Theorem 2.3.3. We replace this with the following argument, using the norm identity (Corollary 5.2).

Since  $u$  is a unit (holomorphic and non-vanishing), the ideals  $\langle g, g' \rangle$  and  $\langle h, h' \rangle$  coincide in the ring of holomorphic functions on  $\Delta'$ :

$$\begin{aligned} g &= uh, & g' &= u'h + uh' \\ \implies h &= u^{-1}g \in \langle g, g' \rangle, & h' &= u^{-1}(g' - u'u^{-1}g) \in \langle g, g' \rangle \\ \implies \langle h, h' \rangle &\subseteq \langle g, g' \rangle. \end{aligned}$$

The reverse inclusion is immediate from  $g = uh$  and  $g' = u'h + uh'$ .

Since  $\tilde{P}(z) \in \langle g, g' \rangle \cap \mathcal{O}(\Delta_2) = \langle h, h' \rangle \cap \mathcal{O}(\Delta_2)$ , Corollary 5.2 gives

$$\tilde{P}(z)^m = \pm \text{disc}(h)(z) \cdot Q(z) \tag{2}$$

for some holomorphic  $Q = N(\bar{b}) \in \mathcal{O}(\Delta_2)$ , where  $m = \deg_{z_r} h$ .

We verify that neither factor on the right is identically zero:

- $\text{disc}(h) \not\equiv 0$  because  $f$  is squarefree, so  $g$  is squarefree as a pseudopolynomial (the coordinate change preserves coprimality of  $g$  and  $g'$ ), and  $h$  divides  $g$  with unit cofactor, so  $h$  is also squarefree.
- $Q \not\equiv 0$  because  $\tilde{P}^m \not\equiv 0$  and  $\text{disc}(h) \not\equiv 0$ , and  $\mathcal{O}(\Delta_2)$  is an integral domain.

Now apply the order additivity lemma (Lemma 4.1) to the restriction of (2) to  $T^* \cap \Delta_2$ :

- $\tilde{P}|_{T^*}$  has constant vanishing order  $\mu$  on  $T^* \cap \Delta_2$  (established in §6.3.3), so  $\tilde{P}^m|_{T^*}$  has constant vanishing order  $m\mu$ .
- $\text{disc}(h)|_{T^*}$  and  $Q|_{T^*}$  are holomorphic, not identically zero, on the connected open set  $T^* \cap \Delta_2 \cong \Delta^{(s)}$ .
- Their product equals  $\pm \tilde{P}^m|_{T^*}$ , which has constant order  $m\mu$ .

By Lemma 4.1,  $\text{disc}(h)|_{T^*}$  has constant vanishing order on  $T^* \cap \Delta_2$ . Denote this constant order by  $r$ . Then  $F(z) := \text{disc}(h)(z) = z_{n-1}^r \cdot N(z)$  (after possibly refining  $\Delta_2$  and noting the power-series structure), where  $N$  is holomorphic and non-vanishing — exactly the hypothesis required by Zariski's theorem.

### 6.3.6 Application of Zariski's theorem (unchanged)

**[UNCHANGED]**

Since  $\text{disc}(h)$  is order-invariant in  $T^* \cap \Delta_2$ , Zariski's 1975 theorem (Theorem 4.1.1) applies to  $h$ : there exists a polydisc  $\Delta_3 \subseteq \Delta_2$  and a holomorphic function  $\psi(z_1, \dots, z_s)$  from  $\Delta_3^{(s)}$  into  $\Delta(0; \epsilon)$  such that for all  $z = (z_1, \dots, z_s, 0, \dots, 0) \in T^* \cap \Delta_3$  and all  $\alpha \in \Delta(0; \epsilon)$ ,

$$h(z, \alpha) = 0 \quad \text{if and only if} \quad \alpha = \psi(z_1, \dots, z_s),$$

and  $h$  is order-invariant in the set  $G^* = \{(z, z_r) : z \in T^* \cap \Delta_3, z_r = \psi(z_1, \dots, z_s)\}$ .

### 6.3.7 Return to the real domain (unchanged)

[UNCHANGED]

The argument on pp. 27–28 of the thesis applies verbatim. In summary:

1. The power series of  $\psi$  has real coefficients (by the Schwarz reflection argument), so  $\eta := \psi|_{B(s)}$  is real-valued and analytic.
2. For  $y \in T \cap B$  and  $\alpha \in (-\epsilon, +\epsilon)$ :  $g(y, \alpha) = 0 \iff h(y, \alpha) = 0 \iff \alpha = \eta(y)$  (using  $u \neq 0$ ).
3. Order-invariance of  $g$  in the graph of  $\eta$  follows from order-invariance of  $h$  in  $G^*$  (since  $g = uh$  and  $u \neq 0$ ).
4. Detransforming via  $\Phi$ : set  $N = \Phi^{-1}(B)$  and  $\theta(x) = (\eta \circ \phi)(x)$ , where  $\phi$  is the chart for  $S$  corresponding to  $\Phi$ . This gives the desired analytic root function and order-invariance.  $\square$

## 7 The Generalized Projection Theorem (Theorem 3.2.3')

[CHANGED — proved from 3.2.1' via the Nullstellensatz]

**Theorem 7.1** (Generalized Projection Theorem, Theorem 3.2.3'). *Let  $A$  be a finite squarefree basis of  $r$ -variate integral polynomials,  $r \geq 2$ , and let  $S$  be a connected submanifold of  $\mathbb{R}^{r-1}$ . Suppose that each element of  $A$  is not identically zero on  $S$ , and each element of  $P'(A)$  (Definition 2.2) is order-invariant in  $S$ . Then:*

1. *Each element of  $A$  is degree-invariant and analytically delineable on  $S$ .*
2. *The sections of  $A$  over  $S$  are pairwise disjoint.*
3. *Each element of  $A$  is order-invariant in every section of  $A$  over  $S$ .*

*Proof.* Let  $F_1, \dots, F_n$  be the nonconstant elements of  $A$ , and set  $f = \prod_{i=1}^n F_i$ . Then  $f$  is squarefree (the  $F_i$  are squarefree and pairwise coprime, so Lemma 7.2 applies).

By hypothesis, we have nonzero order-invariant elements

$$d_i \in \langle F_i, F'_i \rangle \cap \mathbb{R}[x], \quad r_{ij} \in \langle F_i, F_j \rangle \cap \mathbb{R}[x].$$

Set  $R = \prod_i d_i \cdot \prod_{i < j} r_{ij}$ .

**Claim.** Some positive power  $R^N$  belongs to  $\langle f, f' \rangle \cap \mathbb{R}[x]$ .

*Proof of Claim.* Over  $\bar{K}$  (algebraic closure), the repeated-root locus of  $f$  decomposes as

$$V(f, f') = \bigcup_i V(F_i, F'_i) \cup \bigcup_{i < j} V(F_i, F_j).$$

This holds because  $f = \prod F_i$  with the  $F_i$  pairwise coprime and squarefree: a point  $(a, t)$  is a repeated root of  $f$  iff either some  $F_i$  has a repeated root there, or two distinct factors vanish simultaneously.

Since  $d_i \in \langle F_i, F'_i \rangle \cap K[x]$ , the polynomial  $d_i$  vanishes on  $\pi(V(F_i, F'_i))$ . Similarly  $r_{ij}$  vanishes on  $\pi(V(F_i, F_j))$ . Therefore  $R$  vanishes on  $\pi(V(f, f'))$ .

By the Hilbert Nullstellensatz,  $R \in \sqrt{\langle f, f' \rangle \cap \bar{K}[x]}$ , so  $R^N \in \langle f, f' \rangle \cap \bar{K}[x]$  for some  $N \geq 1$ . Since  $f, f'$ , and  $R$  all have integer coefficients, this membership descends to  $\mathbb{R}[x]$  (by flatness, or by a Gröbner basis computation over  $\mathbb{Q}$ ).  $\square$



$R^N$  is a nonzero element of  $\langle f, f' \rangle \cap \mathbb{R}[x]$ . It is order-invariant on  $S$ : each  $d_i$  and  $r_{ij}$  is order-invariant, so  $R$  is order-invariant by Lemma 3.1, and  $\text{ord}_a(R^N) = N \cdot \text{ord}_a(R)$  is constant on  $S$ .

Furthermore,  $f$  is not identically zero on  $S$  and is degree-invariant on  $S$  (the coefficients of each  $F_i$  are order-invariant by hypothesis on  $\text{coeff}(A)$ ).

Applying Theorem 6.1 (Theorem 3.2.1') to  $f$  with  $P = R^N$ :  $f$  is analytically delineable on  $S$  and order-invariant in each  $f$ -section over  $S$ . By Lemma 3.1, each  $F_i$  is order-invariant in every  $f$ -section. The sections of  $A$  over  $S$  are pairwise disjoint (since  $f$  is squarefree and delineable, distinct coprime factors cannot share a root on any section).  $\square$

**Lemma 7.2** (used above). *The product of squarefree pairwise coprime polynomials is squarefree.*

*Proof.* Standard.  $\square$

## 8 Summary of changes from the thesis

| Item                                                               | Status              | Notes                                                                                                     |
|--------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|
| Ch. 2 (all)                                                        | unchanged           | Background material                                                                                       |
| Def. of $P(A)$                                                     | <b>changed</b>      | Generalized to $P'(A)$ with ideal elements                                                                |
| Thm. 3.2.1 statement                                               | <b>changed</b>      | $P \in \langle f, f' \rangle \cap \mathbb{R}[x]$ replaces dis                                             |
| Thm. 3.2.3 statement                                               | <b>changed</b>      | $P'(A)$ replaces $P(A)$                                                                                   |
| Proof of 3.2.1, Assertion 1 from 2                                 | unchanged           |                                                                                                           |
| Proof of 3.2.1, Assertion 2, $s = r - 1$                           | <b>simplified</b>   | $P \neq 0$ directly rules out com                                                                         |
| Proof of 3.2.1, Assertion 2, $1 \leq s \leq r - 2$ : coord. change | unchanged           |                                                                                                           |
| Proof of 3.2.1, $\tilde{P}$ definition                             | <b>changed</b>      | $\tilde{P} = P \circ \Phi^{-1}$ instead of $E = c$                                                        |
| Proof of 3.2.1, complexification                                   | unchanged in method | Same argument applied to $\tilde{P}$                                                                      |
| Proof of 3.2.1, Weierstrass prep                                   | unchanged           |                                                                                                           |
| Proof of 3.2.1, factorization step                                 | <b>changed</b>      | $\tilde{P}^m = \pm \text{disc}(h) \cdot Q$ via no<br>(Cor. 5.2) + order additivity<br>replaces Thm. 2.3.3 |
| Proof of 3.2.1, Zariski application                                | unchanged           |                                                                                                           |
| Proof of 3.2.1, return to real                                     | unchanged           |                                                                                                           |
| Proof of 3.2.3                                                     | <b>changed</b>      | Nullstellensatz argument<br>Thm. 2.3.3 product formula                                                    |
| Ch. 4 (Zariski theorem)                                            | unchanged           | Used as black box                                                                                         |
| <b>New:</b> Order additivity lemma                                 | <b>new</b>          | Lemma 4.1                                                                                                 |
| <b>New:</b> Norm identity for monic polys                          | <b>new</b>          | Lem. 5.1, Cor. 5.2 (standard)                                                                             |

## 9 Status relative to the Lean formalization

The Lean project already contains:

- `GeneralizedProjection.lean`: Theorem 3.2.3' (`mccallum_3_2_3_generalized`), fully proved from the generalized lifting theorem, which is currently an **axiom** (`lifting_theorem_generalized`).
- The Nullstellensatz argument (`nullstellensatz_elim`) is fully formalized.

- All supporting lemmas (order additivity of products, squarefreeness of products, degree-invariance from coefficients, etc.) are formalized.

**Remaining work to remove the axiom:**

1. Formalize the order additivity lemma (Lemma 4.1) for holomorphic functions. This requires Lean infrastructure for holomorphic functions on polydiscs and vanishing order in the complex-analytic setting. *Status:* declared as a standalone axiom (`order_additivity_holomorphic`).
2. Formalize the norm identity (Corollary 5.2). This needs the algebra norm on  $R[z]/(h)$  for monic  $h$  and the identity  $N(\bar{g}) = \pm \text{res}(h, g)$ . The norm map and its multiplicativity are in Mathlib (`Algebra.norm`); the connection to resultants may need to be established.
3. Formalize the proof of Theorem 3.2.1' (this document), using the above. The bulk of the proof (coordinate changes, complexification, Weierstrass preparation, Zariski's theorem, return to real) is shared with the original theorem.

**Note on an earlier incorrect axiom.** An earlier version of the Lean formalization included an axiom `elim_ideal_monik_resultant` asserting  $\langle h, g \rangle \cap R = (\text{res}(h, g))$  for monic  $h$ . This was discovered to be **false** (see Remark in §5) and has been removed. The norm identity (Lemma 5.1) is the correct replacement.